PHO221 · Fall 2026

Physiology & Pathophysiology

Question bank — 142 multiple-choice questions across 6 lectures. Answer key with explanations at the end.

Lecture 1 — Body fluids, membrane transport and water balance

1. Intestinal cells take up glucose against its gradient by coupling it to Na^+ moving down its gradient. This is:
A. Simple diffusion
B. Facilitated diffusion
C. Secondary active transport
D. Primary active transport

2. A patient receives a large volume of a hypotonic solution. What happens to the RBCs?
A. They swell and may lyse
B. They shrink
C. No change
D. They become sickle-shaped

3. After heavy sweating, how does the body restore water balance?
A. ↓ ADH, more water excreted
B. ↑ ADH from the posterior pituitary, more water reabsorbed
C. Kidneys release erythropoietin
D. Thirst is suppressed

4. Why is severe hyperkalemia life-threatening?
A. It causes kidney stones
B. It alters the resting membrane potential, causing arrhythmias
C. It stops ADH release
D. It lowers COP

5. Total body water in an adult male is about what percentage of body weight?
A. 20%
B. 40%
C. 60%
D. 80%

6. Intracellular fluid makes up about what fraction of total body water?
A. 1/3
B. 2/3
C. 1/2
D. 1/4

7. Extracellular fluid is divided mainly into:
A. CSF only
B. ICF and plasma
C. Lymph only
D. Plasma and interstitial fluid

8. The main cation of ICF is:
A. Na^+
B. K^+
C. Ca^{2+}
D. Cl^-

9. The main anion of ECF is:
A. Cl^-
B. Proteins
C. Phosphate
D. K^+

10. Infants have a ___ percentage of body water than adults.
A. Lower
B. Higher
C. Equal
D. Variable only

11. Simple diffusion through the lipid bilayer is fastest for:
A. Large polar molecules
B. Ions
C. Small lipid-soluble molecules (O_2, CO_2)
D. Proteins

12. Facilitated diffusion differs from simple diffusion in that it:
A. Uses a carrier and shows saturation
B. Uses ATP
C. Moves against a gradient
D. Is non-specific

13. Primary active transport uses:
A. Ion gradients
B. Vesicles only
C. Osmosis
D. ATP directly

14. The Na^+/K^+ pump moves:
A. 2 Na^+ out, 3 K^+ in
B. 3 Na^+ out, 2 K^+ in
C. 3 Na^+ in, 2 K^+ out
D. 1 Na^+ out, 1 K^+ in

15. SGLT (Na^+–glucose co-transport) is an example of:
A. Primary active transport
B. Endocytosis
C. Simple diffusion
D. Secondary active transport

16. Na^+/Ca^{2+} exchanger is an example of:
A. Symport
B. Antiport (counter-transport)
C. Uniport
D. Pinocytosis

17. Osmosis is the movement of:
A. Solute to lower concentration
B. Ions in an electric field
C. Water toward higher solute concentration across a semipermeable membrane
D. Proteins

18. A cell placed in a hypotonic solution will:
A. Shrink
B. Stay unchanged
C. Swell and may lyse
D. Crenate

19. 0.9% NaCl is:
A. Hypotonic
B. Hypertonic
C. Isotonic
D. Toxic

20. Phagocytosis is:
A. Cell drinking
B. Diffusion
C. Exocytosis
D. Cell eating of particles

21. ADH conserves water by:
A. Increasing urine volume
B. Increasing water reabsorption in collecting ducts
C. Raising Na^+ excretion
D. Dilating vessels only

22. The thirst centre is in the:
A. Hypothalamus
B. Medulla
C. Cerebellum
D. Cortex

23. Oedema can result from:
A. Raised plasma proteins
B. Low capillary pressure
C. Decreased plasma proteins (low oncotic pressure)
D. High lymph flow

Lecture 2 — Blood and blood disorders

24. In liver failure, low albumin causes oedema because:
A. Colloid osmotic pressure falls, so less fluid returns to capillaries
B. Albumin blocks lymphatics
C. Sodium is pumped into tissues
D. Hydrostatic pressure falls

25. An oxidizing drug converts heme iron from Fe^{2+} to Fe^{3+}. This forms:
A. Oxyhemoglobin
B. Carboxyhemoglobin
C. Methemoglobin
D. Carbaminohemoglobin

26. Pernicious anemia is caused by:
A. Iron deficiency
B. Globin chain defect
C. Lack of intrinsic factor
D. Bone marrow aplasia

27. Low ferritin with high transferrin and small RBCs suggests:
A. B12 deficiency
B. G6PD deficiency
C. Iron deficiency anemia
D. Aplastic anemia

28. G6PD deficiency (favism) is inherited as:
A. Autosomal dominant
B. X-linked
C. Autosomal recessive
D. Mitochondrial

29. Normal RBC life span is:
A. 6–10 days
B. 30–45 days
C. 100–120 days
D. 1 year

30. Normal blood volume in an adult is about:
A. 1–2 L
B. 5–6 L
C. 10 L
D. 15 L

31. Plasma makes up about what percentage of blood?
A. 45%
B. 55%
C. 75%
D. 90%

32. The most abundant plasma protein is:
A. Globulin
B. Fibrinogen
C. Albumin
D. Prothrombin

33. Serum differs from plasma by lacking:
A. Albumin
B. Glucose
C. Water
D. Clotting factors (fibrinogen)

34. Normal haemoglobin in adult males is about:
A. 8–10 g/dL
B. 14–18 g/dL
C. 20–25 g/dL
D. 5 g/dL

35. RBC lifespan is about:
A. 10 days
B. 40 days
C. 120 days
D. 1 year

36. Erythropoietin is produced mainly by the:
A. Liver
B. Kidney
C. Spleen
D. Bone marrow

37. Vitamin B12 or folate deficiency causes:
A. Megaloblastic (macrocytic) anaemia
B. Microcytic anaemia
C. Haemolytic anaemia
D. Aplastic anaemia

38. Iron deficiency anaemia is:
A. Macrocytic hyperchromic
B. Microcytic hypochromic
C. Normocytic
D. Megaloblastic

39. Pernicious anaemia is due to lack of:
A. Intrinsic factor
B. Iron
C. Folate
D. Erythropoietin

40. Sickle cell anaemia results from:
A. Iron loss
B. An abnormal haemoglobin (HbS)
C. Bone marrow failure
D. Blood loss

41. Aplastic anaemia is due to:
A. Haemolysis
B. B12 deficiency
C. Bone marrow failure
D. Bleeding

42. Polycythaemia means:
A. Decreased RBCs
B. Increased RBC count
C. Increased WBCs
D. Low platelets

43. The most abundant WBC is the:
A. Lymphocyte
B. Eosinophil
C. Neutrophil
D. Basophil

44. Eosinophils increase in:
A. Allergy and parasitic infections
B. Bacterial infection
C. Viral infection only
D. Anaemia

45. Platelets are fragments of:
A. Megakaryocytes
B. Lymphocytes
C. RBCs
D. Monocytes

46. Fibrinogen is converted to fibrin by:
A. Plasmin
B. Vitamin K
C. Heparin
D. Thrombin

47. Vitamin K is required for synthesis of factors:
A. I, V, VIII
B. XI, XII
C. II, VII, IX, X
D. III only

48. Haemophilia A is deficiency of factor:
A. VII
B. VIII
C. IX
D. X

49. A person with blood group AB has:
A. Anti-A and anti-B
B. Anti-A only
C. No anti-A or anti-B antibodies
D. Anti-B only

50. The universal donor is group:
A. AB+
B. B−
C. A+
D. O−

51. Haemolytic disease of the newborn occurs when:
A. Rh+ mother carries Rh− fetus
B. Rh− mother sensitized carries Rh+ fetus
C. Both are Rh−
D. Mother is group AB

Lecture 3 — Nervous system I: neurons and action potentials

52. Where is the action potential initiated in a neuron?
A. Dendrites
B. Synaptic terminal
C. Node of Ranvier
D. Axon hillock

53. Which glial cell forms myelin in the CNS?
A. Oligodendrocytes
B. Schwann cells
C. Astrocytes
D. Microglia

54. The hyperpolarization (undershoot) after an action potential is due to:
A. Na^+/K^+ pump reversal
B. Cl^- efflux
C. Na^+ influx
D. Slow closure of voltage-gated K^+ channels

55. During the absolute refractory period no new AP can fire because:
A. K^+ channels are closed
B. The cell is at −90 mV
C. The pump has stopped
D. Voltage-gated Na^+ channels are inactivated

56. Correct order in a spinal reflex arc:
A. Receptor → motor → relay → sensory → effector
B. Receptor → brain → motor → effector
C. Effector → sensory → relay → motor → receptor
D. Receptor → sensory → relay → motor → effector

57. The functional unit of the nervous system is the:
A. Glial cell
B. Synapse
C. Neuron
D. Myelin

58. Myelin in the CNS is formed by:
A. Schwann cells
B. Astrocytes
C. Oligodendrocytes
D. Microglia

59. The resting membrane potential of a neuron is about:
A. +30 mV
B. −70 mV
C. −90 mV only in neurons
D. 0 mV

60. The resting membrane potential is mainly due to:
A. Na^+ influx
B. K^+ efflux through leak channels
C. Ca^{2+} influx
D. Cl^- efflux

61. Depolarization during the action potential is caused by:
A. K^+ efflux
B. Rapid Na^+ influx through voltage-gated channels
C. Cl^- influx
D. Pump activity

62. Repolarization is caused by:
A. Na^+ influx
B. K^+ efflux
C. Ca^{2+} influx
D. Water movement

63. Hyperpolarization after an action potential is due to:
A. Slow closure of K^+ channels
B. Na^+ channel reopening
C. Pump failure
D. Ca^{2+} entry

64. The threshold potential of a neuron is about:
A. −90 mV
B. −55 mV
C. 0 mV
D. +30 mV

65. The all-or-none law means:
A. Larger stimuli give larger action potentials
B. AP size depends on axon length
C. Every stimulus fires
D. A suprathreshold stimulus always produces a full-sized action potential

66. During the absolute refractory period:
A. A stronger stimulus can fire the neuron
B. K^+ channels are closed
C. The neuron is hyperexcitable
D. No stimulus can produce another AP

67. During the relative refractory period:
A. No stimulus works
B. Normal stimuli work
C. A stronger than normal stimulus can produce an AP
D. The membrane is at rest

68. Saltatory conduction occurs in:
A. Unmyelinated fibres
B. Myelinated fibres, jumping between nodes of Ranvier
C. Muscle only
D. Dendrites

69. Conduction velocity increases with:
A. Smaller diameter
B. Lower temperature
C. Larger diameter and myelination
D. Demyelination

70. Local anaesthetics block nerve conduction by blocking:
A. Voltage-gated Na^+ channels
B. K^+ channels
C. Ca^{2+} pumps
D. ACh receptors

71. At a chemical synapse, neurotransmitter release is triggered by:
A. Ca^{2+} influx into the presynaptic terminal
B. Na^+ influx
C. K^+ efflux
D. Cl^- influx

72. An EPSP is caused by:
A. Cl^- influx
B. Hyperpolarization
C. K^+ efflux only
D. Na^+ influx (depolarization)

73. Multiple sclerosis affects:
A. Peripheral myelin
B. CNS myelin
C. Muscle fibres
D. Synaptic vesicles

Lecture 4 — Autonomic nervous system and muscle

74. Postganglionic sympathetic fibres to sweat glands release:
A. Noradrenaline on β_2
B. Dopamine
C. ACh on muscarinic (M3)
D. ACh on nicotinic

75. Parasympathetic stimulation of the eye causes:
A. Mydriasis
B. Dry eye
C. Relaxed ciliary muscle
D. Miosis

76. In myasthenia gravis, antibodies are directed against:
A. Muscarinic receptors
B. Dopamine receptors
C. Acetylcholinesterase
D. Nicotinic ACh receptors at the NMJ

77. In smooth muscle, Ca^{2+} triggers contraction by binding:
A. Troponin C
B. Calmodulin, which activates myosin light-chain kinase
C. Tropomyosin
D. Ryanodine receptors only

78. Parkinson's disease is caused by loss of:
A. Cholinergic neurons
B. Glutamate receptors
C. Myelin
D. Dopaminergic neurons

79. The sympathetic nervous system has a ___ outflow.
A. Craniosacral
B. Thoracolumbar (T1–L2)
C. Cervical
D. Sacral only

80. Preganglionic fibres of both ANS divisions release:
A. Acetylcholine
B. Noradrenaline
C. Dopamine
D. Adrenaline

81. Most sympathetic postganglionic fibres release:
A. Acetylcholine
B. GABA
C. Serotonin
D. Noradrenaline

82. Parasympathetic postganglionic fibres act on:
A. Nicotinic receptors
B. Alpha receptors
C. Muscarinic receptors
D. Beta receptors

83. Sympathetic stimulation causes:
A. Miosis
B. Bradycardia
C. Mydriasis (pupil dilation)
D. Increased gut motility

84. Parasympathetic stimulation of the heart causes:
A. Bradycardia
B. Tachycardia
C. Increased contractility
D. Vasoconstriction

85. Bronchodilation is produced by:
A. Parasympathetic, M3
B. α1
C. Sympathetic, β2
D. Nicotinic

86. The adrenal medulla acts like a modified:
A. Sympathetic ganglion releasing adrenaline into blood
B. Parasympathetic ganglion
C. Motor neuron
D. Sensory receptor

87. Which organ receives only sympathetic innervation?
A. Heart
B. Salivary gland
C. Iris
D. Most blood vessels

88. Skeletal muscle fibres are:
A. Unstriated and involuntary
B. Striated, multinucleated and voluntary
C. Branched with intercalated discs
D. Single-nucleated smooth

89. The neurotransmitter at the neuromuscular junction is:
A. Noradrenaline
B. Dopamine
C. Acetylcholine
D. Glutamate

90. The functional unit of skeletal muscle is the:
A. Sarcomere
B. Myofibril
C. Sarcoplasmic reticulum
D. T-tubule

91. During contraction, which band stays constant?
A. A band
B. I band
C. H zone
D. Sarcomere

92. Calcium triggers skeletal muscle contraction by binding to:
A. Myosin
B. Actin
C. Troponin C
D. Tropomyosin directly

93. Calcium for skeletal muscle contraction comes mainly from:
A. Sarcoplasmic reticulum
B. ECF
C. Mitochondria
D. Nucleus

94. ATP is needed in the cross-bridge cycle for:
A. Actin polymerization
B. Nerve conduction
C. Ca^{2+} release only
D. Detachment of myosin from actin and Ca^{2+} re-uptake

95. Myasthenia gravis is due to:
A. Excess ACh
B. Myelin loss
C. Antibodies against nicotinic ACh receptors
D. Calcium excess

96. Smooth muscle contraction is initiated by Ca^{2+} binding to:
A. Troponin
B. Myosin heads directly
C. Tropomyosin
D. Calmodulin

97. Tetanus (summation) in skeletal muscle results from:
A. Single stimulus
B. Fatigue
C. Low Ca^{2+}
D. High-frequency repeated stimuli

Lecture 5 — Cardiovascular system I: cardiac muscle and potentials

98. The intrinsic firing rate of the SA node is about:
A. 15–40/min
B. 40–60/min
C. 90–105/min
D. 150–200/min

99. Phase 4 of the SA nodal potential is mainly due to:
A. Slow Na^+ influx (funny current)
B. Fast Na^+ influx
C. K^+ efflux
D. Cl^- influx

100. The plateau (phase 2) of the ventricular action potential is due to:
A. Na^+ influx
B. L-type Ca^{2+} influx
C. K^+ influx
D. Na^+/K^+ pump

101. Cardiac muscle cannot be tetanized because:
A. It has no Ca^{2+}
B. It is voluntary
C. Its absolute refractory period is long
D. It lacks gap junctions

102. Cardiac muscle cells are connected electrically through:
A. Gap junctions in intercalated discs
B. Desmosomes only
C. Tight junctions
D. Synapses

103. The resting membrane potential of ventricular muscle is about:
A. −55 mV
B. −70 mV
C. −90 mV
D. 0 mV

104. Phase 0 of the ventricular action potential is due to:
A. Ca^{2+} influx
B. Rapid Na^+ influx
C. K^+ efflux
D. Cl^- influx

105. The plateau (phase 2) of the ventricular action potential is due to:
A. Na^+ influx
B. Slow Ca^{2+} influx balanced by K^+ efflux
C. Cl^- influx
D. Pump only

106. The long refractory period of cardiac muscle prevents:
A. Contraction
B. Relaxation
C. Conduction
D. Tetanus

107. Phase 3 of the ventricular action potential is:
A. Depolarization
B. Resting
C. Plateau
D. Repolarization by K^+ efflux

108. The SA node is the normal pacemaker because it:
A. Has the most negative RMP
B. Has no ion channels
C. Is in the ventricle
D. Has the fastest rate of spontaneous depolarization

109. The pacemaker (prepotential) in SA node is mainly due to:
A. Funny current (If, Na^+ influx) and decreasing K^+ efflux
B. Fast Na^+ channels
C. Cl^- efflux
D. Pump activity

110. Depolarization (phase 0) in SA node cells is due to:
A. Ca^{2+} influx through L-type channels
B. Fast Na^{+} influx
C. K^{+} efflux
D. Cl^{-} influx

111. Sympathetic stimulation increases heart rate by:
A. Slowing the prepotential
B. Hyperpolarizing the SA node
C. Steepening the prepotential slope
D. Blocking Ca^{2+}

112. Vagal stimulation slows heart rate by:
A. Increasing Ca^{2+} influx
B. Releasing noradrenaline
C. Increasing Na^{+} influx
D. Increasing K^{+} permeability (hyperpolarization) and flattening the prepotential

113. Excitation–contraction coupling in cardiac muscle involves:
A. No extracellular Ca^{2+}
B. Calcium-induced calcium release from the SR
C. Only sodium
D. Troponin I alone

114. Hyperkalaemia affects the heart by:
A. Increasing excitability indefinitely
B. Increasing contractility only
C. No effect
D. Depolarizing RMP, which can cause arrhythmia and cardiac arrest

115. Increased extracellular Ca^{2+} tends to:
A. Weaken contraction
B. Increase contractility
C. Stop the SA node
D. Block conduction always

116. The Frank–Starling law states that:
A. Heart rate rises with volume
B. Contraction is independent of stretch
C. Pressure falls with volume
D. Greater end-diastolic volume increases force of contraction

Lecture 6 — Cardiovascular system II: conduction, cycle and blood pressure

117. Where is conduction slowest in the heart?
A. SA node
B. AV node
C. Bundle of His
D. Purkinje fibres

118. EDV = 120 mL and ESV = 50 mL. What is the stroke volume?
A. 50 mL
B. 70 mL
C. 120 mL
D. 170 mL

119. Blood pressure is 120/80 mmHg. The mean arterial pressure is about:
A. 80 mmHg
B. 93 mmHg
C. 100 mmHg
D. 40 mmHg

120. In cardiac muscle, the small Ca^{2+} influx triggers a large release from the SR through:
A. SERCA
B. Calmodulin
C. Ryanodine receptors
D. Na^{+}/Ca^{2+} exchanger

121. The correct order of the conduction system is:
A. AV node → SA node → bundle of His → Purkinje
B. Purkinje → SA → AV
C. SA node → AV node → bundle of His → bundle branches → Purkinje fibres
D. SA → Purkinje → AV

122. The AV nodal delay allows:
A. Ventricles to contract first
B. Atria to empty into ventricles before ventricular contraction
C. Faster heart rate
D. Valve closure

123. The fastest conduction velocity in the heart is in the:
A. AV node
B. SA node
C. Purkinje fibres
D. Atrial muscle

124. The P wave on ECG represents:
A. Ventricular depolarization
B. Atrial repolarization
C. Ventricular repolarization
D. Atrial depolarization

125. The T wave represents:
A. Atrial depolarization
B. Ventricular depolarization
C. AV delay
D. Ventricular repolarization

126. The PR interval mainly reflects:
A. Ventricular contraction
B. Atrial repolarization
C. Conduction through the AV node
D. Ventricular filling

127. Isovolumetric contraction occurs when:
A. All valves are open
B. Only AV valves are open
C. All valves are closed and ventricular pressure rises
D. Blood is ejected

128. The first heart sound (S1) is due to closure of the:
A. Semilunar valves
B. Pulmonary veins
C. Foramen ovale
D. AV (mitral and tricuspid) valves

129. The second heart sound (S2) is due to closure of the:
A. Aortic and pulmonary valves
B. AV valves
C. Chordae
D. Mitral valve only

130. Most ventricular filling occurs during:
A. Atrial systole
B. Ejection
C. Isovolumetric relaxation
D. Rapid passive filling in early diastole

131. Stroke volume is:
A. EDV + ESV
B. HR × EDV
C. EDV − ESV
D. ESV − EDV

132. Cardiac output equals:
A. Blood pressure × resistance
B. Stroke volume × heart rate
C. EDV × ESV
D. Heart rate ÷ stroke volume

133. Ejection fraction is:
A. ESV/EDV
B. HR/SV
C. EDV/SV
D. SV/EDV

134. Mean arterial pressure approximately equals:
A. (SBP + DBP)/2
B. SBP − DBP
C. DBP + 1/3 pulse pressure
D. SBP × DBP

135. Pulse pressure is:
A. Systolic − diastolic
B. Systolic + diastolic
C. Mean pressure
D. Diastolic ÷ 3

136. Blood pressure = cardiac output ×:
A. Heart rate
B. Total peripheral resistance
C. Stroke volume
D. Blood volume

137. The main site of peripheral resistance is the:
A. Arterioles
B. Aorta
C. Capillaries
D. Veins

138. According to Poiseuille's law, resistance is inversely proportional to radius raised to the power:
A. 1
B. 2
C. 3
D. 4

139. Baroreceptors are located in the:
A. Kidney
B. Carotid sinus and aortic arch
C. Hypothalamus
D. Lungs

140. A fall in blood pressure leads the baroreceptor reflex to:
A. Increase vagal tone
B. Dilate arterioles
C. Decrease HR
D. Increase sympathetic activity, raising HR and vasoconstriction

141. Long-term blood pressure control mainly involves:
A. Baroreceptors
B. The cerebellum
C. Chemoreceptors
D. The kidneys and the renin–angiotensin–aldosterone system

142. Angiotensin II raises blood pressure by:
A. Vasoconstriction and aldosterone release
B. Vasodilation
C. Increasing urine output
D. Decreasing ADH

Answer key

Lecture 1

1	C	**Secondary active transport** — The energy comes from the Na^+ gradient (made by the Na^+/K^+ pump), not ATP directly: secondary active co-transport.
2	A	**They swell and may lyse** — Water enters cells in a hypotonic solution, so RBCs swell and can burst.
3	B	↑ **ADH from the posterior pituitary, more water reabsorbed** — High solute concentration is detected by the hypothalamus → ADH release → renal water reabsorption.
4	B	**It alters the resting membrane potential, causing arrhythmias** — K^+ sets the resting potential; disturbing it disrupts cardiac excitability.
5	C	**60%** — Lower in females and the obese (more fat).
6	B	**2/3** — ECF is about 1/3.
7	D	**Plasma and interstitial fluid** — Plasma ≈ 1/4 of ECF, interstitial ≈ 3/4.
8	B	**K^+** — Na^+ dominates ECF.
9	A	**Cl^-** — Plus HCO_3^-.
10	B	**Higher** — About 75%.
11	C	**Small lipid-soluble molecules (O_2, CO_2)** — No carrier or energy needed.
12	A	**Uses a carrier and shows saturation** — E.g. glucose via GLUT.
13	D	**ATP directly** — E.g. Na^+/K^+-ATPase.
14	B	**3 Na^+ out, 2 K^+ in** — Electrogenic, creates negative interior.
15	D	**Secondary active transport** — Uses the Na^+ gradient made by the pump.
16	B	**Antiport (counter-transport)** — Ions move in opposite directions.
17	C	**Water toward higher solute concentration across a semipermeable membrane** — Driven by osmotic pressure.
18	C	**Swell and may lyse** — Water enters.
19	C	**Isotonic** — Normal saline.
20	D	**Cell eating of particles** — Pinocytosis is cell drinking.
21	B	**Increasing water reabsorption in collecting ducts** — Inserts aquaporins.
22	A	**Hypothalamus** — Responds to rising plasma osmolality.
23	C	**Decreased plasma proteins (low oncotic pressure)** — Also raised capillary pressure or lymph obstruction.

Lecture 2

24	A	**Colloid osmotic pressure falls, so less fluid returns to capillaries** — Albumin provides ~80% of COP, the force pulling water back into vessels.
25	C	**Methemoglobin** — Fe^{3+} hemoglobin is methemoglobin, which cannot carry O_2 properly.
26	C	**Lack of intrinsic factor** — Without intrinsic factor from parietal cells, B12 is not absorbed in the ileum → megaloblastic anemia.
27	C	**Iron deficiency anemia** — Iron stores (ferritin) are depleted; transferrin rises to grab more iron. Microcytic picture.
28	B	**X-linked** — It is X-linked. Fava beans, sulfa drugs and antimalarials trigger hemolysis.
29	C	**100–120 days** — RBCs live 100–120 days; platelets only about 6–10 days.
30	B	**5–6 L** — About 8% of body weight.
31	B	**55%** — Cells (mainly RBCs) ~45% = haematocrit.
32	C	**Albumin** — Main contributor to oncotic pressure.
33	D	**Clotting factors (fibrinogen)** — Serum is obtained after clotting.
34	B	**14–18 g/dL** — Females ~12–16 g/dL.
35	C	**120 days** — Destroyed mainly in the spleen.
36	B	**Kidney** — Stimulated by hypoxia.
37	A	**Megaloblastic (macrocytic) anaemia** — Impaired DNA synthesis.

38	B	**Microcytic hypochromic** — Small pale RBCs.
39	A	**Intrinsic factor** — Needed for B12 absorption.
40	B	**An abnormal haemoglobin (HbS)** — A haemolytic anaemia.
41	C	**Bone marrow failure** — All cell lines may fall (pancytopenia).
42	B	**Increased RBC count** — Raises viscosity.
43	C	**Neutrophil** — ~60–70%, first line against bacteria.
44	A	**Allergy and parasitic infections** — Eosinophilia.
45	A	**Megakaryocytes** — Count 150,000–400,000/µL.
46	D	**Thrombin** — Final step of coagulation.
47	C	**II, VII, IX, X** — Warfarin antagonizes vitamin K.
48	B	**VIII** — Haemophilia B = factor IX.
49	C	**No anti-A or anti-B antibodies** — Universal recipient.
50	D	**O−** — No A, B or Rh antigens.
51	B	**Rh− mother sensitized carries Rh+ fetus** — Prevented with anti-D.

Lecture 3

52	D	**Axon hillock** — The axon hillock has the lowest threshold and generates the impulse.
53	A	**Oligodendrocytes** — Oligodendrocytes myelinate CNS axons; Schwann cells do it in the PNS.
54	D	**Slow closure of voltage-gated K^+ channels** — K^+ keeps leaving while K^+ channels close slowly, so the membrane dips below rest.
55	D	**Voltage-gated Na^+ channels are inactivated** — Inactivated Na^+ channels cannot reopen until the membrane repolarizes.
56	D	**Receptor → sensory → relay → motor → effector** — Sensory (afferent) in, relay neuron in the cord, motor (efferent) out to the muscle.
57	C	**Neuron** — Dendrites receive, axon transmits.
58	C	**Oligodendrocytes** — Schwann cells in the PNS.
59	B	**−70 mV** — Inside negative.
60	B	**K^+ efflux through leak channels** — Plus the Na^+/K^+ pump contribution.
61	B	**Rapid Na^+ influx through voltage-gated channels** — Reaching threshold opens Na^+ channels.
62	B	**K^+ efflux** — Voltage-gated K^+ channels open.
63	A	**Slow closure of K^+ channels** — Excess K^+ leaves.
64	B	**−55 mV** — Must be reached to fire.
65	D	**A suprathreshold stimulus always produces a full-sized action potential** — Subthreshold stimuli produce no AP.
66	D	**No stimulus can produce another AP** — Na^+ channels are inactivated.
67	C	**A stronger than normal stimulus can produce an AP** — Coincides with hyperpolarization.
68	B	**Myelinated fibres, jumping between nodes of Ranvier** — Faster and more energy-efficient.
69	C	**Larger diameter and myelination** — Type A fibres are fastest.
70	A	**Voltage-gated Na^+ channels** — No depolarization.
71	A	**Ca^{2+} influx into the presynaptic terminal** — Vesicle exocytosis.
72	D	**Na^+ influx (depolarization)** — IPSP: Cl^- influx or K^+ efflux.
73	B	**CNS myelin** — Slows or blocks conduction.

Lecture 4

74	C	**ACh on muscarinic (M3)** — Sweat glands are the exception: sympathetic but cholinergic (ACh → M3).
75	D	**Miosis** — It contracts the sphincter pupillae (miosis) and the ciliary muscle for near vision.
76	D	**Nicotinic ACh receptors at the NMJ** — Blocked nicotinic receptors stop transmission → muscle weakness; anticholinesterases help.
77	B	**Calmodulin, which activates myosin light-chain kinase** — Ca^{2+}–calmodulin activates MLCK, which phosphorylates myosin light chains.

78	D	**Dopaminergic neurons** — Death of dopaminergic neurons controlling movement → tremor, bradykinesia.
79	B	**Thoracolumbar (T1–L2)** — Parasympathetic is craniosacral.
80	A	**Acetylcholine** — Acting on nicotinic receptors.
81	D	**Noradrenaline** — Exception: sweat glands (ACh).
82	C	**Muscarinic receptors** — Released ACh.
83	C	**Mydriasis (pupil dilation)** — Fight-or-flight.
84	A	**Bradycardia** — Via vagus nerve, M2 receptors.
85	C	**Sympathetic, β2** — Basis of salbutamol in asthma.
86	A	**Sympathetic ganglion releasing adrenaline into blood** — ~80% adrenaline.
87	D	**Most blood vessels** — Also sweat glands, adrenal medulla.
88	B	**Striated, multinucleated and voluntary** — Cardiac muscle has intercalated discs.
89	C	**Acetylcholine** — On nicotinic (Nm) receptors.
90	A	**Sarcomere** — Z line to Z line.
91	A	**A band** — I band and H zone shorten.
92	C	**Troponin C** — Tropomyosin moves to expose myosin-binding sites.
93	A	**Sarcoplasmic reticulum** — Released after T-tubule depolarization.
94	D	**Detachment of myosin from actin and Ca^{2+} re-uptake** — Lack of ATP → rigor mortis.
95	C	**Antibodies against nicotinic ACh receptors** — Treated with anticholinesterases.
96	D	**Calmodulin** — Activates myosin light-chain kinase.
97	D	**High-frequency repeated stimuli** — Twitches fuse.

Lecture 5

98	C	**90–105/min** — Vagal tone slows it to around 75/min at rest.
99	A	**Slow Na^+ influx (funny current)** — The funny current (If) slowly depolarizes the membrane toward threshold.
100	B	**L-type Ca^{2+} influx** — Ca^{2+} entry through L-type channels balances K^+ efflux and holds the plateau.
101	C	**Its absolute refractory period is long** — The long absolute refractory period lasts through systole and early diastole.
102	A	**Gap junctions in intercalated discs** — Acts as a functional syncytium.
103	C	**−90 mV** — More negative than SA node.
104	B	**Rapid Na^+ influx** — Fast depolarization.
105	B	**Slow Ca^{2+} influx balanced by K^+ efflux** — L-type Ca^{2+} channels.
106	D	**Tetanus** — Allows ventricular filling.
107	D	**Repolarization by K^+ efflux** — Ca^{2+} channels close.
108	D	**Has the fastest rate of spontaneous depolarization** — ~70–80 beats/min.
109	A	**Funny current (If, Na^+ influx) and decreasing K^+ efflux** — Then T-type Ca^{2+} channels.
110	A	**Ca^{2+} influx through L-type channels** — Slow upstroke.
111	C	**Steepening the prepotential slope** — β1 → more If and Ca^{2+} current.
112	D	**Increasing K^+ permeability (hyperpolarization) and flattening the prepotential** — ACh on M2 receptors.
113	B	**Calcium-induced calcium release from the SR** — Trigger Ca^{2+} enters during the plateau.
114	D	**Depolarizing RMP, which can cause arrhythmia and cardiac arrest** — Inactivates fast Na^+ channels.
115	B	**Increase contractility** — More trigger calcium.
116	D	**Greater end-diastolic volume increases force of contraction** — Intrinsic regulation of stroke volume.

Lecture 6

117	B	**AV node** — About 0.05 m/s at the AVN, giving the ventricles time to fill.
118	B	**70 mL** — SV = EDV − ESV = 120 − 50 = 70 mL.
119	B	**93 mmHg** — MAP = DBP + ⅓ pulse pressure = 80 + 40/3 ≈ 93 mmHg.

120	C	**Ryanodine receptors** — Calcium-induced calcium release via ryanodine receptors. SERCA pumps it back for relaxation.
121	C	**SA node → AV node → bundle of His → bundle branches → Purkinje fibres** — Normal sequence.
122	B	**Atria to empty into ventricles before ventricular contraction** — ~0.1 s.
123	C	**Purkinje fibres** — ~4 m/s; AV node slowest.
124	D	**Atrial depolarization** — QRS = ventricular depolarization.
125	D	**Ventricular repolarization** — Atrial repolarization is hidden in the QRS.
126	C	**Conduction through the AV node** — Normal 0.12–0.20 s.
127	C	**All valves are closed and ventricular pressure rises** — Begins with AV valve closure.
128	D	**AV (mitral and tricuspid) valves** — Start of systole.
129	A	**Aortic and pulmonary valves** — Start of diastole.
130	D	**Rapid passive filling in early diastole** — Atrial contraction adds ~20–30%.
131	C	**EDV − ESV** — ~70 mL at rest.
132	B	**Stroke volume × heart rate** — ~5 L/min at rest.
133	D	**SV/EDV** — Normal ~55–70%.
134	C	**DBP + 1/3 pulse pressure** — Diastole lasts longer.
135	A	**Systolic − diastolic** — Normally ~40 mmHg.
136	B	**Total peripheral resistance** — BP = CO × TPR.
137	A	**Arterioles** — Resistance vessels.
138	D	**4** — Halving radius raises resistance 16×.
139	B	**Carotid sinus and aortic arch** — Short-term BP control.
140	D	**Increase sympathetic activity, raising HR and vasoconstriction** — Restores pressure.
141	D	**The kidneys and the renin–angiotensin–aldosterone system** — Regulates blood volume.
142	A	**Vasoconstriction and aldosterone release** — ACE inhibitors block its formation.